Problem 12 | Problems T6andT7_Electric and Magnetic Properties.pdf #12 | Microstructure · ferromagnetic properties

Effect of Grain Size, Impurities and Porosity on a Ferromagnetic Material

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12. Specify in the following table how the increase in the following microstructural parameters affect the listed properties of a ferromagnetic material

	Grain size	***Impurities***	***Porosity***
Hysteresis losses			
Conductivity			
Permeability			
Coercitivity			
Saturation Magnetization			
Curie Temperature			

The sheet prints no solution for this problem.

HYPOTHESES & PARAMETERS

Hypotheses

1. **One parameter at a time.** Each column is an increase of that parameter with the composition of the ferromagnetic material and the other parameters kept constant.
2. **Domain walls control the loop.** A larger grain gives the Bloch walls more mobility, and inclusions, dislocations and pores reduce it. With a larger mobility the permeability grows and the coercivity and the remanence fall; with a smaller mobility the opposite happens [Session 15 Slide 21].
3. **Impurities and porosity as obstacles.** The slide groups inclusions, dislocations and pores as obstacles to the walls, so impurities (as particles of a second phase) and pores act in the same way on the loop. The effect of dissolved atoms is the same in direction (extension).
4. **Intrinsic properties.** The saturation magnetization is constant for a material of a given composition [Session 15 Slide 20]. The Curie temperature is the temperature at which the saturation magnetization falls to zero, and the table of Slide 23 gives a different value for each element; both are fixed by the composition and the bonding, not by the microstructure (the link between T_c and the composition is an extension).
5. **Conductivity.** The electrical resistivity grows with the structural imperfections (grain boundaries, dislocations) and with the impurities [Session 13 Slides 16-17]; the effect of pores is an extension.

6. **Hysteresis losses.** The energy lost per cycle is the area of the loop (extension, as in Problem - T7-11 Hysteresis Loops of Soft and Hard Magnetic Materials): a wider loop (higher H_c) means higher losses.

Unknown: the effect (increase, decrease or none) of each parameter on each of the six properties.

MODEL & STRATEGY

Two chains separate the properties. The first goes from the microstructure to the mobility of the Bloch walls, and from the mobility to the permeability, the coercivity and the area of the loop. The second contains the properties that the microstructure does not change at all: the saturation magnetization and the Curie temperature, set by the composition. The conductivity follows its own chain, the scattering of the conduction electrons.

Parameter	Mobility of the Bloch walls	Permeability	Coercivity	Loop area (losses)
Larger grain size	increases	increases	decreases	decreases
More impurities, pores	decreases	decreases	increases	increases

Strategy.

1. Fill the first chain from Slide 21.
2. Decide the conductivity from the scattering of the electrons, with Session 13.
3. State why the saturation magnetization and the Curie temperature do not change with the microstructure.
4. Collect the answers in the table of the statement.

STEP-BY-STEP CALCULATION

1. Grain size [Session 15 Slide 21]

A larger grain size increases the mobility of the Bloch walls, because there are fewer grain boundaries to stop them. So:

- permeability μ: **increases**;
- coercivity H_c: **decreases** (and the remanence B_r too);
- hysteresis losses: **decrease**, since a narrower loop has a smaller area;
- conductivity: **increases**, because the grain boundaries are structural imperfections that scatter the electrons and raise the resistivity [Session 13 Slides 16-17].

2. Impurities [Session 15 Slide 21]

Inclusions of another phase are obstacles that pin the walls, and the mobility decreases. So:

- permeability: **decreases**;
- coercivity: **increases** (and the remanence too);

- hysteresis losses: **increase**;
- conductivity: **decreases**, because foreign atoms and particles raise the residual resistivity ρ_i [Session 13 Slide 16].

3. Porosity [Session 15 Slide 21]

Pores are listed on the slide with the inclusions, so the loop changes as for the impurities:

- permeability: **decreases**;
- coercivity: **increases**;
- hysteresis losses: **increase**;
- conductivity: **decreases** (extension): the pores remove conducting section and scatter the electrons, so the same direction as for the impurities.

4. Saturation magnetization [Session 15 Slide 20]

The saturation magnetization is constant for a material of a given composition: it is reached when all the dipoles of the solid are aligned, and the number and the intensity of the dipoles are fixed by the composition. Therefore:

- grain size: **no effect**;
- impurities: **no effect** while they are only inclusions; if an impurity dissolves and changes the composition the value changes (extension);
- porosity: **no effect** on the material itself. Per unit of total volume the pores only dilute the magnetization (extension).

5. Curie temperature [Session 15 Slide 23]

T_c is the temperature at which the saturation magnetization falls abruptly to zero. Each material has its own value (Fe 1043 K, Co 1404 K, Ni 631 K, Gd 289 K), which depends on the elements, not on the size of the grains or on the pores. Grain size and porosity have **no effect**. Impurities have **no effect** as inclusions; dissolved elements that change the composition can shift it (extension).

6. Result

Property	Grain size increases	Impurities increase	Porosity increases
Hysteresis losses	decrease	increase	increase
Conductivity	increases	decreases	decreases
Permeability	increases	decreases	decreases
Coercivity	decreases	increases	increases
Saturation magnetization	no effect	no effect as inclusions	no effect
Curie temperature	no effect	no effect as inclusions	no effect

RESULT & CHECKS

Checks

- **Consistency of the table.** Permeability and coercivity move in opposite directions in every column, as the slide says (μ up with H_c and B_r down for a larger grain, and the reverse for the obstacles). The losses follow the coercivity.
- **Impurities and porosity act alike.** Slide 21 puts inclusions, dislocations and pores in the same line, and the table gives the same entries for both columns, except for the properties that depend on the composition.
- **Limits.** For no obstacles (a pure, dense, coarse-grained material) the walls move freely: the highest permeability and the lowest coercivity and losses. For a very porous material the loop widens and the section that carries the flux and the current shrinks.

Interpretation

One rule explains the table: whatever stops the Bloch walls raises the coercivity and the losses and lowers the permeability, and whatever frees them does the opposite. The saturation magnetization and the Curie temperature are properties of the composition, so no change in the microstructure alone moves them. The same rule is used on purpose in the two classes of Problem T7-11: soft magnetic materials are made pure, dense and with large grains, and hard magnetic materials are made with obstacles (fine grains, precipitates), which is an extension of the slide. A higher conductivity also raises the currents induced in a core under alternating fields, which adds to the losses; this is an extension, and the table asks only for the effects listed.

Check against the official solution

Item	Official	Computed	Comment
All 18 entries of the table	not printed on the sheet	table of Section 3	permeability, coercivity: Session 15 Slide 21; saturation magnetization: Slide 20; conductivity: Session 13 Slides 16-17; the entries marked extension (porosity and conductivity, effect of composition on M_s and T_c, losses as loop area) are not printed on the slides